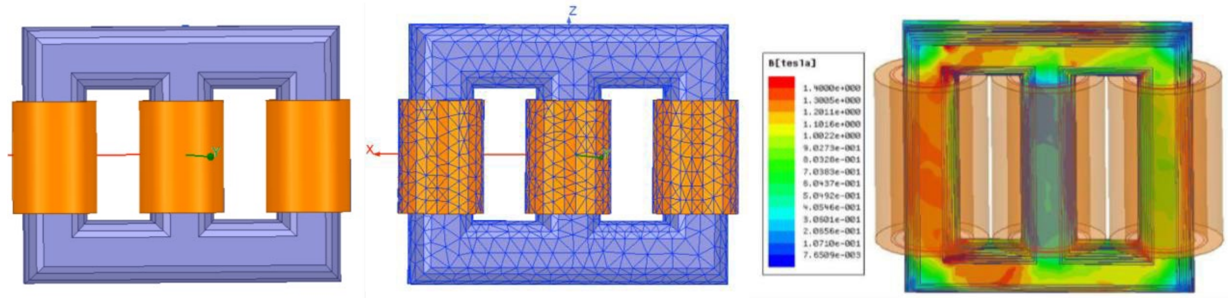
The University of New South Wales

ELEC3115 – Electromagnetic Engineering

PRELAB 1: PERMITTIVITY CONSTANT (ε) OF DIELECTRIC MATERIALS

1. Introduction

Finite Element Analysis (FEA) is a numerical method used to solve complex engineering problems that involve three-dimensional partial differential equations. Instead of solving the whole system at once, FEA breaks it into many small elements and analyses how each element behaves under given conditions. Together, these results approximate the behaviour of the entire system.



(a) (b) (c)

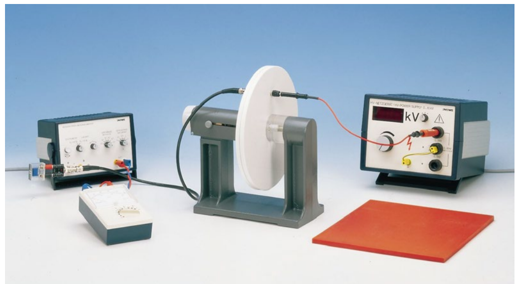
Fig. 1. FEA modelling of a power transformer: (a) CAD model, (b) Meshed model broken down into small elements, and (d) Solved magnetic flux distribution on the model.

In electromagnetics, FEA is used to analyse devices such as motors, transformers, and generators. It allows us to visualise fields—such as electric or magnetic fields—inside complicated structures. In Figure 1, a transformer model is divided into small elements, and the resulting magnetic field is displayed after solving.

In this Prelab, you will apply the same idea to an **electrostatic** problem involving a parallel-plate capacitor.

2. Overview of Prelab 1

Your task is to draw, set up, and solve the electric field between two parallel copper plates using **ANSYS Maxwell 2D**. The plates form a parallel-plate capacitor with cylindrical symmetry about the **Z-axis**. Because the plate spacing is small, a 2D cross-section is sufficient for this analysis. Figure 2 shows the 2D geometry you will recreate in Maxwell. The lab setup that you will use during the laboratory session is also shown.



Lab setup of a parallel plate capacitor

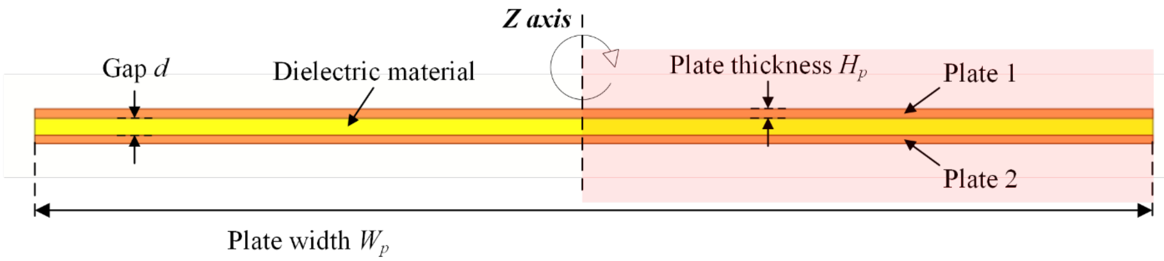


Fig. 2. Cross-section of the parallel plate capacitor.

3. Objectives

In this Prelab, you will:

- build a 2D model of a parallel-plate capacitor,
- assign materials and voltage excitations,
- apply boundary conditions,
- generate a mesh suitable for the student software limit, and
- run an electrostatic simulation to study the electric field and capacitance. (see appendix)
- write a 2-page report by answering a set of questions (see below)

4. Pre-lab Tasks

Task 1: Model setup and field visualisation

Follow the provided video instructions (and/or simplified instructions in the Appendix) to build a **parallel plate capacitor model** in ANSYS Maxwell (Electrostatic solver).

After solving the model:

- Generate a **vector plot of the electric field** in the region between and around the plates.
- Ensure the plot clearly shows the field behaviour **between the plates and near the edges**.
- Take a **clear, well-labelled screenshot** of this plot.
- Include the screenshot in your 2-page pre-lab report.

Be prepared to **briefly explain your modelling steps and interpret the field plot orally** to the demonstrator at the start of the lab.

Task 2: Analysis and Discussion

Answer the following questions concisely in your 2-page pre-lab report. Clearly label each answer. Focus on **correct results, clear reasoning, and physical interpretation**, as you will be asked to explain these orally.

Question 1: Capacitance – theoretical vs FEA
Using the equations introduced in the lecture, calculate the **theoretical capacitance** of the parallel plate capacitor using the given geometry and material properties.

- Clearly state any assumptions used in the theoretical model.
- Obtain the **capacitance from the FEA simulation**.
- Compare the two values and comment on any difference, referring to modelling assumptions and electric field behaviour.

Question 2: Comparison with experimental result (Post-lab)
For a plate separation of **d = 2 mm**:

- Compare the **theoretical** and **FEA-calculated** capacitance values with the **experimentally measured capacitance**.
- Identify which value is closer to the experimental result and briefly explain why.

Question 3: Fringing (edge) effect
From the electric field vector plot, observe the field behaviour near the plate edges.

- Describe the **fringing (edge) effect** observed in the simulation.
- Explain how this effect leads to a difference between the **ideal theoretical capacitance** and the **FEA-calculated capacitance**.

Question 4: Charge density near plate edges
The surface charge density plot shows a sharp increase near the outer edges of the plates.

- Explain why charge accumulates near the edges.
- Relate this behaviour to the local electric field distribution.

Question 5: Total charge estimation from FEA
Using the **FEA-calculated surface charge density**:

- Estimate the **total charge on one plate**.
- You may neglect the very high peak values near the plate edges.
- Briefly describe the method used for this estimation.

Question 6: Comparison of charge values (Post-lab)
Compare the **FEA-estimated total charge** with the **theoretical** and **experimental** charge values.

- Comment on the level of agreement between these values.
- Briefly explain the main sources of difference.

5. Marking rubric (10 marks)

- **Correct simulation setup and results (6 marks)**
  - Correct model geometry, boundary conditions, and solution (3)
  - Correct field plots and extracted quantities (capacitance/charge) (2)
  - Clear presentation in a concise 2-page report (1)
- **Oral explanation and interpretation (4 marks)**
  - Clear explanation of modelling steps (2)
  - Correct physical interpretation of results (field behaviour, edge effects, discrepancies) (2)

Appendix: ANSYS Maxwell Modelling of the parallel plate capacitor

This simplified version of the instructions is designed to help you follow the steps more easily while maintaining technical accuracy. These instructions should be used in conjunction with the instruction video.

1. Starting ANSYS Maxwell

(How to access ANSYS, see the separate instruction document)

1. Open **ANSYS Electronics Desktop** using either:
   - the desktop icon, or
   - the program menu (via myAccess or local installation: refer to installation guide).
2. Create a **2D Maxwell Project**:
   - Select **Maxwell → Maxwell 2D**.
   - Rename the default project to **Prelab1_Plate_Capacitor**, by right clicking the default name or from the Edit menu.
3. Set the **Solution Type** to **Electrostatic**.by right clicking the file name or from the Maxwell 2D menu
4. Under **Geometry Mode**, choose **Cylindrical about Z**.

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2. Drawing the Parallel Plate Capacitor

2.1 Draw the Plates

1. **Plate 1**
   - Draw → *Rectangle*.
   - Start at coordinates **(0, 0,0)**.
   - Set **dx = 130.1 mm**,**dy =0**, **dz = 1 mm**.
   - Rename it **Plate1** in properties window
2. **Plate 2**
   - Edit>Select All
   - Edit>Duplicate>Along Line
   - Start at **coordinate (0,0,0)**
   - **Set dx=0, dy=0, dz=3 mm**
   - In Duplicate Along Line Window
     1. Set Total number: 2
     2. Do not check Attach to Original object
   - Rename the new rectangle **Plate2**.

You may have to use 'Fit All' view to see both plates clearly.

2.3 Draw the Dielectric

1. Draw a rectangle between the two plates as shown in the video.
2. Name it **Dielectric** and set its colour to yellow.
3. You can also change the plates colour to orange and make them somewhat transparent.

2.2 Draw the Boundary Region

1. Draw → *Rectangle*.
2. Start at coordinate **(0, 0, -15)**.
3. Set **dx =165**, **dy =0, dz=34**.
4. Rename it **Boundary**.
5. Set **Boundary** to Wireframe

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3. Assigning Materials

Assign materials as follows:

- **Plate1**, **Plate2** → *Copper*
- **BND**, **Dielectric** → *Vacuum* (you may later change the dielectric material)

To change the dielectric material:

1. Select **Dielectric** in the history tree.
2. In the **Properties Window**, click **Edit…** under *Material*.
3. Search for a material (e.g., *air*, *PVC plastic*).
4. Select and click **OK**.

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4. Setting Voltage Excitations

1. Select **Plate1** → Maxwell 2D → Excitations → Assign → *Voltage*.
   - Name: **Ground**
   - Value: **0 V**
2. Select **Plate2** → Assign → *Voltage*.
   - Name: **Voltage1**
   - Value: **Vdc × 1 V**
   - Create variable **Vdc = 1000**.

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5. Capacitance Computation

1. Go to **Maxwell2D** → **Parameters** → **Assign** → **Matrix**.
2. Select:

- **Ground** → *Ground*
- **Voltage1** → *Signal*

3. Click **OK**.

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6. Mesh Settings

The free student version of ANSYS limits the mesh to **2000 elements**, so a medium-fineness mesh is used.

6.1 Plates and Boundary

1. Select **Plate1**, **Plate2**, and **BND**.
2. Maxwell 2D → Mesh → Assign Mesh Operation → *Length Based*.
3. Set:
   - **Max element length = 5 mm**

6.2 Dielectric

1. Select **Dielectric**.
2. Assign another *Length-Based* mesh.
3. Set:
   - **Max element length = 2 mm**

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7. Running the Simulation

1. **Validation Check**
   - Maxwell 2D → *Validation Check*
   - Ensure all items are marked as valid.
2. **Solve**
   - Maxwell 2D → *Analyze All*.

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8. Solution

1. Results → *Solution Data* (to check numerical value of the capacitance)

2. Plot *Field Vectors*

   - Edit > Select Objects > All Model Objects
   - Maxwell 2D → Field > E > E_vector
   - In Volume > AllObject

3. For a better view of field vectors, double click legend window:

   - Marker/Arrow tab > size 1.5e-2
   - Plot tab> Min:0.6, Max:10,

- Adjust the spacing slider for the best view of the edging effect at the end of the two plates.

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9 How to change the value of d (size of dielectric)

In Model window: Select > Dielectric

- Change zSize to *d*. (You have now made Dielectric’s depth a design variable)
- A window will pop up to input the value. First, use the initial value as shown in the instructions. Once you set *it here, it will be available under design properties to change to other values,* like the change of voltage Vs values.
